FEDERAL SPECIFICATION

TAPES, MEASURING (STEEL, SURVEYORS')

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. CLASSIFICATION

1.1 Types. — Surveyors' steel measuring tapes covered by this specification shall be furnished in the types specified in the invitation for bids, using the following six essential elements: (1) description of ribbon (2) length (nominal) of graduated interval, (3) width, (4) thickness of ribbon from which tape is made, (5) location of end points, and (6) type of graduation. (See 6.1, 6.2, and 6.3 for suggestions in specifying these six elements.)

2. APPLICABLE SPECIFICATIONS

2.1 There are no other specifications applicable to this specification except those referenced in section 7 by individual departments.

(Copies of Federal specifications and the Federal Specifications Index may be obtained upon application, accompanied by money order, coupon, or cash, to the Superintendent of Documents, Government Printing Office, Washington 25, D. C., or to the GSA Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo, Dallas, Denver, San Francisco, and Seattle. Prices may be obtained from the Superintendent of Documents or from the GSA Regional Offices.)

3. REQUIREMENTS

3.1 Material.—Material used throughout shall be of best quality normally used in good commercial practice.

3.2 Identification. — Each tape shall be plainly and permanently marked with the manufacturer's name or with a trade-mark of such known character that the source of manufacture may be readily determined.

3.3 Graduations and figures. — Graduations and figures shall be permanent and distinct. Graduation marks shall be of such relative lengths that the value is readily determined. When an etching process is used, the total reduction in thickness at any point, as a result of the etching, shall not exceed 40 percent of the original thickness.

3.4 End loops.—Unless otherwise specified, each tape shall be furnished with end loops and two leather thongs.

3.5 Reels.—When reels are required, they shall be as specified in the invitation for bids.

3.6 Accuracy. — The error in the total length of the tape, when supported horizontally throughout its length at the standard temperature of 68° F. (20° C.) and at standard tension, shall be not more than 0.1 inch per 100 feet (2 millimeters per 25 meters). The standard tension is 10 pounds (4.5 kilograms) for tapes 25 to 100 feet or from 10 to 30 meters in length, and 20 pounds (9 kilograms) for tapes longer than 100 feet or 30 meters. All similar intervals between graduations shall be the same length.

3.7 Workmanship. — Workmanship shall be first class in every respect. Tapes shall be free from defects which might affect the durability and serviceability.

Source: https://assist.dla.mil -- Downloaded: 202640-26T11:48Z
Check the source to verify that this is the currenversion before use.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.—One or more sample tapes of each type, from each lot delivered shall be taken by the inspector, and the samples taken shall be considered as representative of the whole lot of the same type.

4.2 Inspection. — The samples shall be inspected to determine compliance with the requirements of this specification.

4.3 Test.

4.3.1 *Accuracy.*—At least one tape of each type shall be taken from the sample tapes and tested for the accuracy requirements specified in 3.6.

5. PREPARATION FOR DELIVERY

5.1 Packaging. — Unless otherwise specified in the invitation for bids, commercial packaging is acceptable under this specification.

5.2 Packing. — The subject commodity shall be prepared for shipment to permit acceptance by carrier for transportation at the lowest applicable rate, and to afford maximum protection from normal hazards of transportation.

5.3 Marking.

5.3.1 *Packages.*—Each package shall be plainly marked to indicate the name and classification of the commodity (see 1.1 and 6.1), and the quantity contained therein, as defined by the contract or order under which shipment is made, and the name of the manufacturer.

5.3.2 *Shipping containers.*—Unless otherwise specified in the invitation for bids, shipping containers shall be marked with the name and classification of the commodity (see 1.1 and 6.1), and the quantity contained therein, as defined by the contract or order under which shipment is made, the name of the contractor, and the number of the contract or order.

6. NOTES

6.1 The following items may be useful in specifying the type of tape to be furnished:

6.1.1 *Description of ribbon:*

1. Black background with bright markings in relief.
2. Black background with main divisions indicated by intaglio markings on integral bright bosses.
3. Chrome background with black markings.
4. White background with black markings in relief.
5. Corrosion - resistant background with babbitt metal bosses.
6. Corrosion - resistant background with brass or nickel-silver sleeves.
7. Bright steel with brass or nickel-silver sleeves.

6.1.2 *Length (nominal) of graduated interval:*

1. 100-foot
2. 200-foot
3. 300-foot
4. 1-chain
5. 2-chain
6. 5-chain
7. 8-chain
8. 30-meter
9. 50-meter

6.1.3 *Width:*

1. 1/8-inch
2. 3/16-inch
3. 1/4-inch
4. 5/16-inch
5. 3/8-inch

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

6.1.4 *Thickness of ribbon from which tape is made:*

1. 0.008-inch
2. 0.010-inch
3. 0.012-inch
4. 0.015-inch
5. 0.020-inch
6. 0.025-inch

} ± 10 percent

and for tapes with babbitt metal bosses,

7. 0.016 inch ± 15 percent

6.1.5 *Location of first graduation:*

1. Approximately 6 inches from end of tape.
2. At end of tape.

6.1.6 *Type of graduation:*

1. Feet, tenths, and hundredths throughout.
2. Feet, with first and last foot to tenths and hundredths.
3. Feet, with first and last foot to tenths.
4. Feet, with extra foot before zero to tenths and hundredths numbered from zero.
5. Feet, with an extra foot before zero to tenths and hundredths numbered from zero, and with last foot to tenths and hundredths.
6. Feet, with extra foot before zero to tenths numbered from zero.
7. Feet, inches, and eighths.
8. Links, with first and last link to tenths.
9. Links (except to 10 links only beyond 100 links).
10. Links, with first link to tenths, but to 10 links only beyond 100 links.
11. Links, with first 10 links to tenths, balance of first 100 links, but to 10 links only beyond 100 links.
12. Meters, decimeters, and centimeters, with first decimeter to millimeters.
13. Meters and decimeters, with first and last decimeter to millimeters, and with balance of first and last meter to centimeters.
14. Meters and decimeters, with extra decimeters before zero to millimeters numbered from zero.
15. Meters, decimeters, and centimeters, with first and last decimeters to centimeters and millimeters.
16. Meters, decimeters, and centimeters, with extra decimeter before zero to millimeters numbered from zero.

6.2 Tapes graduated on both sides may be ordered when specified in the invitation for bids.

6.3 Not all combinations of the items listed in 6.1 and 6.2 are available or even feasible. Before requesting bids or ordering tapes under this specification, it is suggested that manufacturers' catalogs be consulted to determine what combinations are available.

6.4 For tapes in cases for general use see Federal Specification GGG–T–106, Tapes; Measuring, General-Use.

6.5 It is believed that this specification adequately describes the characteristics necessary to secure the desired commodity and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.6 Federal specifications do not include all types, classes, grades, sizes, etc., of the commodities indicated by the titles of the specifications, or which are commercially available, but are intended to cover the types, etc., which are suitable for Federal Government requirements.

Source: https://assist.dla.mil -- Downloaded: 202640-26T11:48Z
Check the source to verify that this is the current version before use.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

7. DEPARTMENTAL REQUIREMENTS

7.1 The following departmental specifications and other publications, of the issues in effect on date of invitation for bids, and special requirements, form a part of this specification for purchases made under this specification by the respective departments.

7.2 Army and Air Force.

7.2.1 *Applicable specifications and other publications.*

U. S. Army Specification:

100–2—Standard Specification for Marking Shipments by Contractors.

Corps of Engineers Tentative Specification:

T–2483—Processing and Packing for Overseas Shipment of Instruments, Measuring—Basic Specification for.

(Copies of specifications, standards, and drawings required by the contractor in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

7.2.2 *Army purchases.*

7.2.2.1 *Overseas shipment.*—Tape shall be preserved, packaged, and packed in accordance with Corps of Engineers Specification T–2483.

7.2.2.2 *Marking.* — Shipping containers shall be marked in accordance with U. S. Army Specification 100–2.

7.3 Navy.

7.3.1 *Applicable specifications and other publications.*

Federal Specifications:

NN–B–591—Boxes, Fiberboard, Wood-Cleated (for Domestic Shipment).

NN–B–601—Boxes, Wood-Cleated-Plywood, for Domestic Shipment.

NN–B–621—Boxes, Wood, Nailed and Lock-Corner.

NN–B–631—Boxes; Wood, Wirebound (for Domestic Shipment).

LLL–B–631—Boxes; Fiber, Corrugated (for Domestic Shipment).

LLL–B–636—Boxes; Fiber, Solid (for Domestic Shipment).

Military Specifications:

JAN–P–105—Packaging and Packing for Overseas Shipment — Boxes, Wood, Cleated, Plywood.

JAN–P–106—Packaging and Packing for Overseas Shipment — Boxes; Wood, Nailed.

JAN–B–107—Boxes, Wood, Wirebound (Overseas Type).

JAN–P–108—Packaging and Packing for Overseas Shipment — Boxes, Fiberboard (V-Board and W-Board), Exterior and Interior.

JAN–P–116—Packaging and Packing for Overseas Shipment — Preservation, Methods of.

JAN–P–125—Packaging and Packing for Overseas Shipment—Barrier-Materials, Waterproof, Flexible.

JAN–P–138—Packaging and Packing for Overseas Shipment — Boxes, Wood, Fiberboard-Lined.

JAN–P–140—Packaging and Packing for Overseas Shipment — Adhesives, Water-Resistant, Case-Liner.

Navy Department Specification:

General Specifications for Inspection of Material.

Bureau of Supplies and Accounts Publication:

Navy Shipment Marking Handbook.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

(Copies of specifications, standards, and drawings required by the contractor in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

7.3.2 *Inspection procedures (Navy only).*— For Naval purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

7.3.3 *Packaging.* — Tapes shall be individually packaged, without preservative, in accordance with the requirements of Method IA of Military Specification JAN–P–116.

7.3.4 *Packing.*

7.3.4.1 *For domestic shipment.* — Unless otherwise specified, tapes, packaged as described in 7.3.3, shall be packed in wood-cleated fiberboard boxes, wood-cleated plywood boxes, nailed wood boxes, wirebound wood boxes, corrugated fiber boxes, or solid fiber boxes conforming to the requirements of Specification NN–B–591, NN–B–601, NN–B–621, NN–B–631, LLL–B–631 or LLL–B–636. The gross weight of wood boxes shall not exceed approximately 200 pounds except where the weight of an individual item exceeds this amount. The gross weight of fiberboard boxes shall be subject to the limitation of the box specification.

7.3.4.2 *For overseas shipment.* — Tapes, packaged as described in 7.3.3, shall be packed in cleated-plywood boxes, nailed wood boxes, wirebound wood boxes, fiberboard boxes, or fiberboard-lined wood boxes, conforming to the requirements of Military Specification JAN–P–105, JAN–P–106, JAN–B–107, JAN–P–108 or JAN–P–138. The gross weight of wood boxes shall not exceed approximately 200 pounds except where the weight of an individual item exceeds this amount and of fiberboard boxes, approximately 70 pounds.

7.3.4.2.1 Contents shall be enclosed within a sealed waterproof bag, or its equivalent, made from material conforming to the requirements of Military Specification JAN–P–125, for case-liner material. Seams and closures shall be sealed with adhesive conforming to the requirements of Military Specification JAN–P–140.

7.3.4 *Marking.*

7.3.5.1 *Interior packages.* — Each interior package shall be durably and legibly marked with the following information:

Material description
Supplier or manufacturer
Contract or order number
Fed. Spec. GG–T–116
Quantity contained
Date packed
Service stock number

7.3.5.2 *Shipping containers.* — In addition to any special marking required by the contract or order, shipments shall be in accordance with the requirements of the Navy Shipment Marking Handbook.

7.3.6 *Ordering data.* — Requests, requisitions, schedules, and contracts or orders should specify whether the subject commodity is to be packed and shipping containers marked for domestic or overseas shipment. (See 7.3.4 and 7.3.5.)

NAVY INTEREST: A*MC* S Y

This specification is a part of Section IV, Part 5, of the Federal Standard Stock Catalog.

For sale by the Superintendent of Documents, Washington 25, D. C. Price 5 cents.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.